

Historic Glass Tutorial

Monticello Field School 2026

Written by Iris Puryear, 6/2025. Last updated Miranda Leclerc, 6/2026.

Material/Composition

- **Silica:** silicon dioxide, most often in the crystalline form of sand
 - Melting point is 1713°C (3115°F), not ideal for industrial production
- **Flux** or **alkali:** lowers melting point of silica and facilitates fusion of glass
 - Melting point is 1400-1600°C (2552-2912°F)
 - Most common fluxes are soda ash (sodium carbonate, typically in the form of marine plant ash) and potash (potassium carbonate, typically in the form of wood ash)
- **Stabilizer:** material added to glass mixture to make stronger and less brittle
 - Most common stabilizers are lime (calcium oxide) and lead (lead oxide)
- Most commonly encountered historic glass formulas
 - **Soda Lime Glass** = silica + soda ash + lime
 - **Potash Lead Glass** = silica + potash + lead
 - *Can identify lead content in material by ice-blue color of fluorescence under short wave UV light*

Manufacturing Technique

- **Free Blown**
 - Refers to vessels that are produced entirely by hand, without the use of molds or machinery
 - Key attributes
 - Asymmetry
 - No evidence for mold seams, molded decoration or lettering
 - Surfaces generally smooth and glossy
 - Curvature of body typically more curvilinear with no sharp angles
 - Globular and elongated cylinders most common
 - Glass more evenly distributed throughout various parts of the vessel when compared with mold blown examples
 - The base of the vessel will have a pontil mark or scar where the pontil rod was removed after shaping the finish
 - The appearance of the pontil mark depends on the technique used to attach the pontil to the object. On some objects, the residue/pontil mark was ground down or heat polished, especially on tablewares.

**absence of mold seam or presence of pontil scar alone does not indicate a manufacturing technique*

- **Mold Blown**

- Refers to vessels shaped by forcing molten glass into the outline of a mold through air pressure applied by mouth through a blow pipe
 - Vessel shoulders, neck and base may also be formed in the mold, or may be finished by hand and tooled
 - Mold may have one or multiple constituent parts
 - After vessel is removed from mold, the **finish must always still be hand-manipulated or tooled in some way**
 - *Note: Technically, molds are also used for machine-made glass, and for this reason, some material culture researchers consider “machine-made” as a sub-category of “mold blown.” At Monticello, however, we distinguish between machine-molded and hand-molded glass for dating purposes. We use the term “mold blown” only in reference to historic, non-machine-made vessels that have been formed in a mold.*
- Key attributes
 - Presence of mold seams* (=raised lines on vessel surface caused by hot glass forced into the edges of a hinged mold)
 - Presence of molded lettering and/or molded decoration
 - Vessel shape more regular and symmetrical relative to free blown examples
 - Body shape may have sharp angles and can be multi-sided or paneled (e.g. square, octagonal, etc.)
 - The more decorated an object or more complex a shape, the more clever makers get with hiding mold seams
 - May or may not have a pontil scar at the base
 - Textured surface
 - Mold interior often imparts an “orange peel” like texture to exterior surface of vessel
 - **Location of mold seams is most important diagnostic**
 - Mold seams do **not** run all the way up the vessel, and **never** extend onto or over the lip
 - Seams usually disappear at the shoulders, along the neck, or at the base of the finish
 - → This is because the top of the vessel is reheated when the glass maker forms the finish, and this process typically erases any mold lines in this area
 - **Caveat is that molded vessels do not always have visible mold seams. 1-part “dip” molds, for example, typically do not have mold seams because the mold has no hinges and therefore no gaps where seams can form. Seams can also be removed via other processes, such as marvering and fire polishing.*

Identifying decoration can be a clue to manufacturing technique and mold type

**you will not be quizzed on decoration or mold type for this field school, but it is good knowledge to have in your archaeologist's toolkit*

- **Contact Molding:** Molten glass is blown into a full-sized mold to produce a full-sized vessel
 - Full-sized mold can be one-piece or multi-part
 - Key attribute here is that the interior and exterior surfaces of a contact molded vessel have a concave-convex relationship—that is, if the curvature of one surface is convex, the opposite surface will have a corresponding concavity in the same place
 - Most common type of mold, esp. for bottle/container glass, both historically and into modern era
- **Pattern Molding:** Glass is first blown into a part-sized mold with a decorative pattern imprinted on the surface, then vessel is removed from the smaller mold and free blown to full size
 - Key attribute is that interior and exterior surfaces mirror one another— a ridge on the outside will always be paired with a corresponding ridge on the inside of the vessel
 - Pattern often becomes distorted or diffuse from blowing the vessel to full size, i.e., vessel shape is not formed in the mold
 - Typically only found on tablewares and flasks, but occasionally used for other container types
 - Technology exists since Roman times, but pattern-molded vessels become increasingly rare from the mid-19th century onward
- **Optic Molding:** Glass is first blown into a part-sized mold with a pattern imprinted on the surface, then transferred to a full-sized mold and blown to full size
 - Essentially a combination of pattern and contact molding
 - The two-part process transfers the patterned design of the first, part-sized mold onto the *interior* surface of the full-sized vessel
 - The *exterior* surface of the finished vessel takes on the configurations of the second, full-sized mold (typically undecorated)
 - Relatively uncommon—generally only found on tablewares dating from the 18th century or later

- **Press Molding:** Pressure is applied with a plunger to force molten glass to conform to the outlines of a mold
 - Because interior surface is shaped by the plunger, it is smooth and does not necessarily follow the curvature of the vessel exterior
 - Exterior surface is shaped by the mold, and is often decorated with a pattern
 - Typically used for open-mouthed vessels such as tablewares and jars because process requires a wide enough opening at the top of vessel to remove the plunger
 - Most common in archaeological contexts from the early 19th onward (although technology exists earlier)
 - Technically, the glass is not literally “blown” into a press mold, in that the pressure is supplied by plunger (not air pressure) – but archaeologists generally still consider press molding a subcategory of “mold blown” glassware

Note: *With the exception of pattern molding, all of the above molding techniques were eventually translated to the machine manufacturing process*

- **Mouth Blown**
 - Generalized term referring to glass vessels formed by air pressure that is applied through a blowpipe by the mouth (i.e., not by a machine).
 - Applies to fragments that lack any clear indicators of free blown or mold blown manufacture—but which are clearly not machine made
 - Most often applied to historic wine bottle glass or dark green bottle glass (DGBG) or colorless leaded/unleaded glass tablewares
- **Machine-Made**
 - Similar process to mold blowing, except that the air pressure or power that forces the glass into the shape of the mold is now supplied by machine, or by a mechanized plunger (in the case of machine press molding)
 - Semi-automatic and fully automatic machines appear late 19th/early 20th century
 - In semi-automatic, glass gobs are hand fed into the machine by a person and in the case of the 1901 Owens patent machine, also had to be hand-finished
 - In automatic glass machines, glass gobs are parsed by the machine
 - Basic process
 - First, a quantity of molten glass enters a combination of two molds: (1) a “ring” or finish mold and (2) a part-sized or parison mold
 - Ring mold shapes the finish to full size, while the body is partially formed in the part-sized mold
 - Part-sized mold is removed from the bottom and the partially formed vessel (still attached to the ring mold at the top) is placed in a second, full-sized mold where it is blown to full size

- Key attributes
 - *Note that while many attributes of machine-made glass (e.g., mold seams, symmetry) overlap with mold blown examples, the following diagnostics are exclusively found on machine-made vessels*
 - Again, **location of mold seams is important diagnostic**
 - Mold seams should **go over the finish and up to or over the lip** of the vessel (*caveat is that mold seams are sometimes removed via fire-polishing*)
 - Ghost Seams: Faint, irregular or “wandering” seam lines that appear parallel to the main, more prominent mold seam on some machine-made bottles
 - Owen’s Suction Scar: Circular mark on base of vessel with “feathering” edges, often off-center. Result of shearing off the glass.
 - Valve Marks: Regular circular mark or indentation on base of vessel, typically around ½ inch in diameter. Typically indicative of press and blow manufacture.
 - Stippling/Knurling: Textured surface seen on the base of many machine-made bottles (post-1940)

Form

- **Bottle/Container**
 - Wine-Style Bottle
 - Generalized term used to describe globular or cylindrical shaped bottles manufactured from dark green or olive-green glass
 - Vessel shape can be used as approximate dating mechanism
 - → In general, vessels become less globular, taller and more cylindrical over the course of the 17th and 18th centuries (*see Noel Hume pp. 60-71 for detailed typology*)
 - Pharmaceutical Bottle
 - Generalized term used to describe a LARGE variety of bottles which held a variety of liquids and medicines, including proprietary/patent medicines, syrups, tonics, or other products sold at chemists and apothecaries
 - Often “smaller” (not always the case) or “thinner” (than alternative glass types like soda bottles or wine bottles, but this is not a hard and fast rule)
 - Manufactured in wide variety of molds and finishes, can be decorated or undecorated
 - May be branded with maker or product name
 - Other common bottle forms include case, mineral/soda, beer/liquor, condiment/food, cosmetic, ink, etc.
 - Jars
 - Mason Jar patented 1858

- Very common on archaeological sites from the late 19th century onward
- **Tableware**
 - Stemware
 - “Stemware is a general term used for glass vessels that have a foot and a bowl joined by a stem.” (JefPat, 2010)
 - Often in the form of stemmed glasses, primarily drinking vessels such as wine glasses.
 - Used for a variety of purposes, could also be part of dinner or dessert services, serving dishes. Function often indicated by size and shape of stemware.
 - Stem, bowl, and foot shape can assist in dating
 - Tumbler
 - Flat bottomed, handle-less, and cylindrical open-mouthed drinking vessel
 - Very common on archeological sites
 - Can have a combination of manufacturing and decorative techniques
 - Additional forms include decanters, salvers, salts, dessert glasses, cruets/castors, bowls, plates and cups
 - Often manufactured of leaded (potash-lead) glass, but important to look for additional attributes like vessel shape and decoration to ID
 - Often has engraved, molded, cut, or other decoration
 - Manufacture often difficult to pin down for tablewares
 - Multiple different manufacturing techniques often used on the same vessel
 - Finishing techniques frequently used to erase diagnostic mold lines and smooth out pontil scars

Other Glass (Non-Vessel) Forms

- Window Glass
- Plate Glass
 - Refers to larger category of thick, cast, flat glass used for a variety of purposes, including mirrors and furniture inlay
- Lamp chimney
- Other forms of lighting glass proliferate in the 19th century (e.g., lamp globes and shades, oil lamps, lantern glass, etc.)

Supplementary Materials

(fear not – you shall not be tested on this)

- **Important 19th Century Advances in Molded Glassware**

- Snap case (1857)
 - Case or clamp device that replaces role of pontil rod for holding bottles while neck/lip are finished
 - Technology eliminates pontil mark on vessel bases
 - This another reason why the absence of pontil mark doesn't necessarily mean a vessel is machine-made—vessels produced with snap case lack pontil mark, but finish still must be hand tooled
- Lipping tool (c. early-mid 19th century)
 - Handheld tool that clamps onto bottle neck and forms a cleaner, more regular finish on mold blown vessels (almost looks machine-made)
 - Process removes all extant mold seams around area of finish
- Press mold
 - First press mold patented by Bakewell in Philadelphia for making furniture knobs in 1825
 - First patent for machine made pressed glassware in 1826 by New England Glass Company (side lever machine)
- In the 1860s, innovations in refining the raw materials to produce glass results in a lime glass recipe that was strong, refractive, workable, and set slowly- all the preferred qualities of lead glass at about 1/3 the price. By the 1870s, pressed lead glass is largely phased out by lime glass.

- **20th Century Tips**

- Applied color labels: production of baked-on enamel labels begins 1934
- “FEDERAL LAW PROHIBITS SALE OR REUSE OF THIS BOTTLE” → Bottles with this mark date from 1935-1964
- Stippling/knurling on bottle bases post-dates 1940
- Pure Food and Drug Act of 1906 requires volume disclosure on bottle or label (i.e. ‘8 oz’ or ‘ONE PINT’)

- **Glass Color**

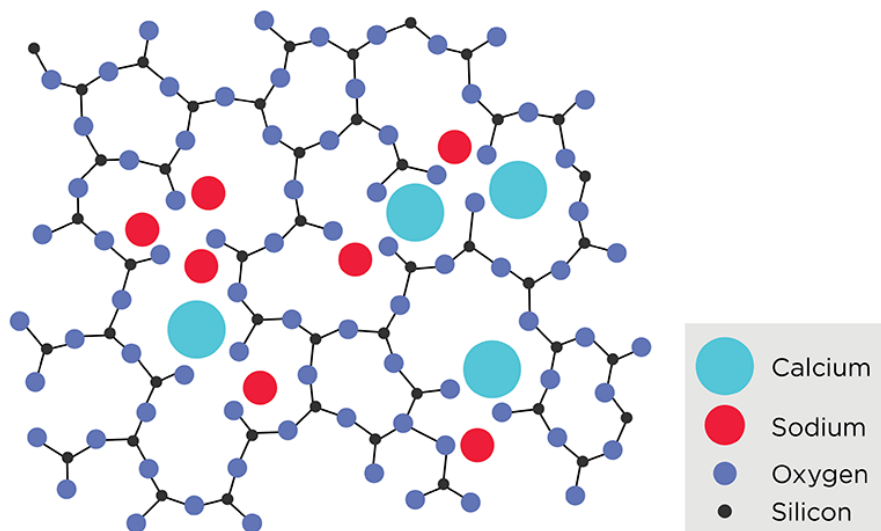
- Glass color can sometimes be useful as a dating tool and in identifying vessel form (albeit with caveats)

- Colorless/Clear
 - Colorless glass requires adding a decolorizing agent to counteract impurities in glass that naturally impart color to the material
 - Most common decolorizing agents historically are manganese and lead oxide
 - Because the technology for colorless glass is complex and historically challenging to manufacture, it was largely restricted to more expensive tableware forms
 - Venetians mastered recipe for colorless soda lime glass by at least the 15th century, which they achieved by adding small amount of manganese oxide, but this technology was not widely distributed
 - English don't master technology for colorless glass until invention of potash-lead glass in late 17th century
 - Following invention of leaded glass, soda lime glass is largely replaced by potash-lead glass for most tableware forms between the late 17th through 19th centuries
 - By the end of the 19th century, however, more advanced recipes for colorless soda lime glass are marketed and begin to appear again on tableware forms
 - Soda lime eventually replaces potash lead as most common type of glass used for tablewares in the 20th century
 - Manganese dioxide was commonly used in early 20th century as decolorizing agent on many container forms
 - Manganese turns glass a light purple when exposed to sunlight (i.e., "solarized" glass)
- Dark Green/Olive Green
 - Although there are always exceptions, dark green/olive green colored glass is almost always associated with bottle glass, especially wine-style or case bottle
- Amber/Brown
 - As a general rule of thumb, lighter colored amber glass typically dates to the 19th century, while darker amber or brown ("beer bottle brown") bottle glass is often machine-made
 - *However, it is often difficult to distinguish nuanced colors at the fragment level, and vessel thickness plays a significant role in how light or dark the glass color appears*
- "Sprite Green"
 - Very light, neon green color most closely associated with modern Sprite bottles
 - This color occurs almost exclusively on machine-made bottles

- **Non-Molded Glass Decorative Techniques** (non-exhaustive, what we commonly see on tablewares at Monticello)
 - Cut (abrasive): The technique whereby glass is removed from the surface of an object by grinding and polishing it with rotating wheels made of stone, wood, and cork.
 - Attributes to look for: sharp edges, relatively deep incisions, inconsistent depth/width/length
 - Engraved (abrasive): when a design is scratched or stippled onto the glass surface using a sharp implement and an abrasive, predominately a diamond point or copper wheel. Creates a frosted, rough surface.
 - Attributed to look for: recessed and frosted surface, under magnification will see stippling, scratching, line drawing and scribbly-ness
 - See more decorative techniques in Jones, 1985 and JefPat Table Glass webpage

How glass works (for the material science nerds)

- Silica + flux is unstable and brittle.
 - When cooled and hardened, glass is an amorphous solid. On the molecular level, some of the molecules are arranged as a crystalline solid and some are arranged randomly as if a liquid.
 - Alkalis are soluble salts, so dissolve in water. Potassium and sodium are alkaline/basic, which means they tend to shed electrons. Shedding electrons attract water.
 - The glass absorbs moisture, pollutants, and is vulnerable to fluctuations in temperature or other environmental factors which causes deterioration known as glass rot, crizzling, or glass disease.
 - Weeping, crazing, devitrification, delamination, etching or frostiness, pitting/spalling, crumbling
- Alkalis work by disrupting the bonds between silicon dioxide molecules ► less heat needed to melt
- Stabilizers function by bonding to unbonded oxygen atoms in the silica, as if plugging the 'holes' left in the silica structure created by the alkalis. Stabilizers balance out the charge of unbonded atoms in glass and reduce reactivity to environmental factors like H₂O. Essentially, closes the gaps between silica and contains the fluxes.



Molecular structure of soda lime glass (Kopp Glass)

Recommended Resources

- Jones, O. (Olive R.), Sullivan, C., Parks Canada. National Historic Parks and Sites Branch. (1985). *The Parks Canada glass glossary for the description of containers, tableware, flat glass, and closures*. Ottawa, Ont.: National Historic Parks and Sites Branch, Parks Canada, Environment Canada
 - Digital copy here → <https://sha.org/bottle/pdf/GlassGlossary.pdf>
- <https://apps.jefpat.maryland.gov/diagnostic/TableGlass/index-TableGlass.html>
- <https://sha.org/bottle/>
- Window glass manufacture demo: <https://www.youtube.com/watch?v=kzBXU2ovfGo>
- Anything from the Corning Museum: <https://allaboutglass.cmog.org/>
- Pressed glass: <https://youtu.be/cWxtpqxEgaI?si=YP5mZ5CrkAUmQmcv\;>
https://youtu.be/oJlQyOTpQmU?si=a_WYdAgMPzQyqHA1
- Owen bottle machine: <https://bottleinfo.historicbottles.com/pdf/michaelowens.mpg>; 4:38 for footage <https://youtu.be/Ifa0U9vF6ds?si=F8hc-xfhQRKyAjo6>
- Copper wheel engraving: <https://youtu.be/3C1mOf3Z85I?si=WUStQBcTuO3Pa3E6;>
<https://youtu.be/DEWzIJGUjfQ?si=2I4H958jnVlCfjHC>
- Virtual Museum of Historic Bottles and Glass: <https://fohbcvirtualmuseum.org/>
- JefPat table glass website: <https://apps.jefpat.maryland.gov/diagnostic/TableGlass/index-TableGlass.html>
- A fun trade catalog: https://cmog.primo.exlibrisgroup.com/discovery/delivery/01CORNING_INST:01CORNING_INST/1259862540004126